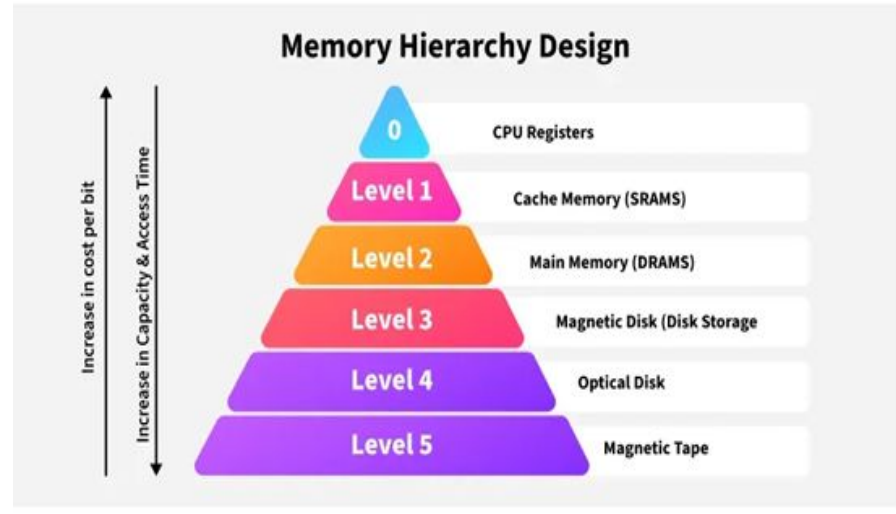

Memory

K.LOKESH

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Memory Hierarchy

- A memory hierarchy organizes different types of memory in a computer system based on access speed, cost, and capacity.
- It aims to balance speed and cost by providing fast but smaller memory closer to the CPU and slower but larger memory further away.
- This arrangement allows for efficient data access, as frequently used data can be stored in faster memory levels, while less frequently used data can be stored in slower, larger, and more cost-effective levels.



Levels of Memory Hierarchy

1. **CPU Registers** The fastest and smallest memory, directly accessed by the CPU, often used for current operations and data.
2. **Cache Memory** Faster than main memory but smaller in capacity, used to store frequently accessed data and instructions for faster retrieval by the CPU.
3. **Main Memory (RAM)** Larger and slower than cache memory, the primary storage for active programs and data, readily accessible by the CPU.
4. **Secondary Storage (Hard Disks, SSDs)** Slower than main memory but with high capacity, used for storing data that is not currently in use but needs to be preserved.
5. **Tertiary Storage (Magnetic Tapes, Optical Disks)** The slowest and largest memory, used for archival storage of data that is rarely accessed.

Classification Of Memory

- **Volatile Memory**
 - Loses data when power is off.
- **Registers**
 - Located inside the CPU Store immediate data like operands, counters
 - Extremely fast (access time in nanoseconds)
 - Very limited in size (typically 8–32 general-purpose registers)
- **Cache Memory**
 - Small memory between CPU and RAM
 - Stores frequently accessed data/instructions
 - Types:
 - L1 Cache: Closest to CPU, smallest and fastest
 - L2 Cache: Slightly larger and slower
 - L3 Cache: Shared among CPU cores, larger but slower

Classification Of Memory

- **RAM**
 - Used for storing runtime data like variables, stack, heap, buffers.
 - SRAM (Static RAM)
 - DRAM (Dynamic RAM)
- **Non-Volatile Memory**
 - Retains data without power, common types are:
 - 1. ROM (Pre-programmed)
 - 2. Flash Memory (USB, Microcontrollers)
 - 3. Secondary Memory (HDD, SSD)

Cache Memory

- Memory that is smaller and faster than RAM is called cache memory.
- It is a volatile memory fixed closer to the CPU to provide high-speed data access to the processor and stores frequently used computer programs, applications and data.
- Generally Access Me

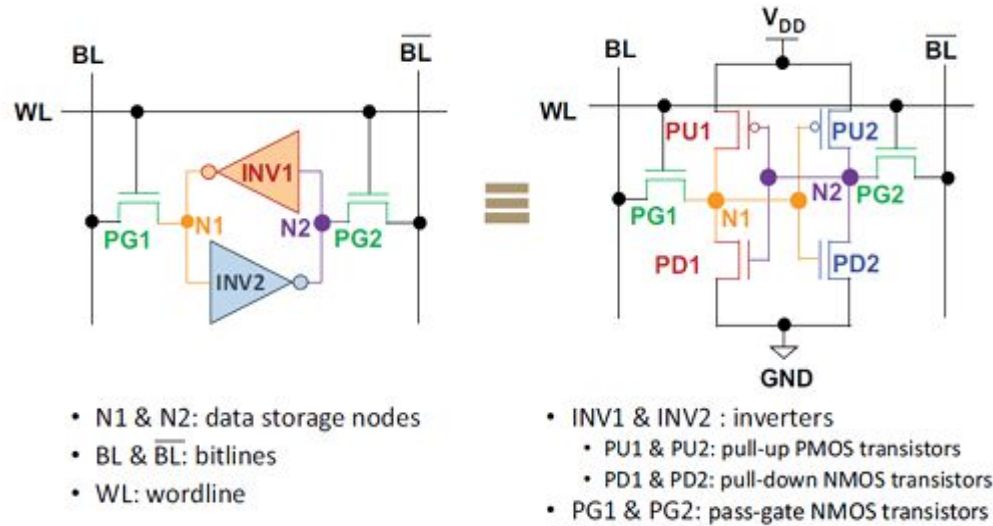


or and Random

SRAM

- **Static Random-Access Memory (SRAM)** is a type of semiconductor memory that uses bistable latching circuitry to store each bit of data.
- Static means data is retained as long as power is supplied, without the need for refreshing.
- Random access means that each bit of the data could be read/written independently.
- Used for fast-access, low-latency applications.
- Requires no refresh like DRAM because it uses flip-flop circuits.
- Larger in size. (Low density device)
- There are two kinds of power: Static and Dynamic Power. In SRAM:
 - Static power consumption = High (Due to leakage in the always-on transistors).
 - Dynamic power consumption = Low (There's no refresh as DRAM)
- Commonly used in:
 - CPU caches (L1, L2, L3)
 - Register files
 - On-chip buffers

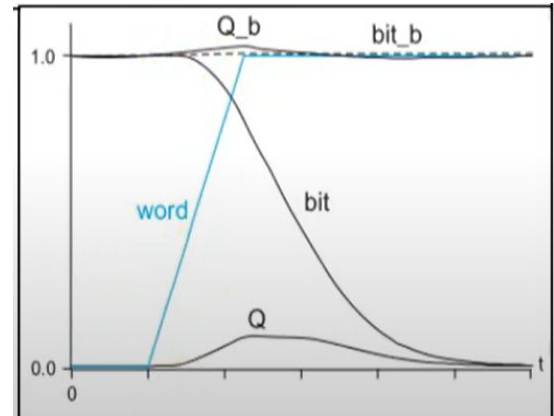
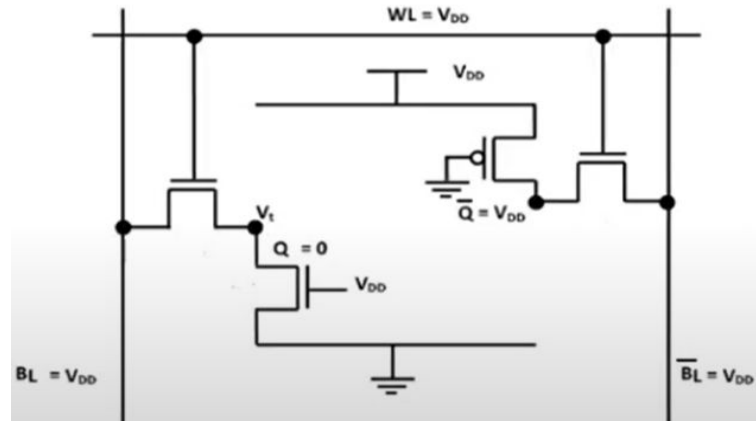
6T SRAM Cell Design



- The core of the 6T cell includes:
 - INV1 and INV2: Two cross-coupled inverters i.e Complementary storage nodes.
 - N1 and N2: These are two storage nodes of the SRAM cell, as the data are stored at one of these two nodes.
 - BL and $\overline{BL}$ (Bit Lines): The storage nodes are connected to BL and $\overline{BL}$ through two access transistors, whose gates are controlled by the same wordline (WL).
 - WL(Word Line): Controls access to the cell.

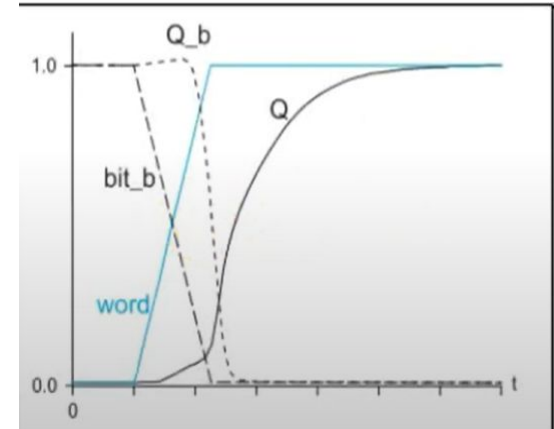
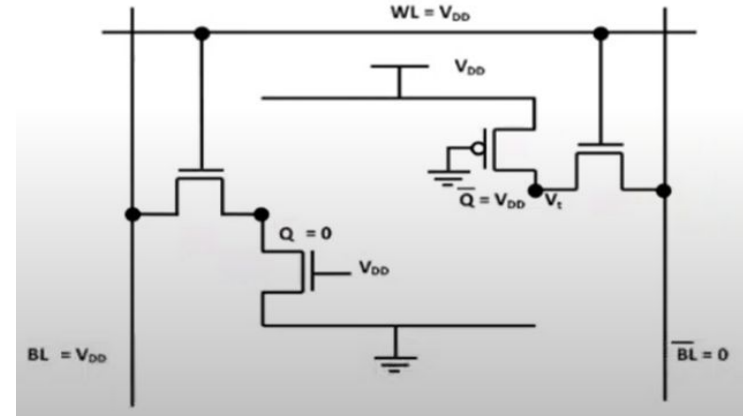
SRAM Read Operation

- Assume "0" at N1 and "1" at N2
- The drivers first recharge all the bitlines (BL and BL) to VDD (logical 1 value)
- $N1 = 0$ and $N2 = 1$, BL start discharging and BL stay at high.
- A small voltage difference (e.g., 100 mV) builds up between BL and BL.
- Now sense amplifier comes in picture. A sense amplifier compares the two and rapidly drives the output to logic level:
 - If $BL > BL$ Output = 1
 - If $BL < BL$ Output = 0



SRAM Write Operation

- Bit Lines (BL and $\overline{BL}$) are driven externally with the value to be written. Write drivers pull BL and $\overline{BL}$ to the desired logic levels.
- Word Line Activation : WL goes high, turning ON access transistors. This connects the internal cell nodes N1 and N2 to BL and $\overline{BL}$
- Cell Overwrite: The strong write drivers force the voltage onto N1 and N2.
 - If we are writing '1': N1 = 1, N2 = 0
 - If we are writing '0': N1 = 0, N2 = 1
 - The previous value stored is overwritten by overpowering the cross-coupled inverter.



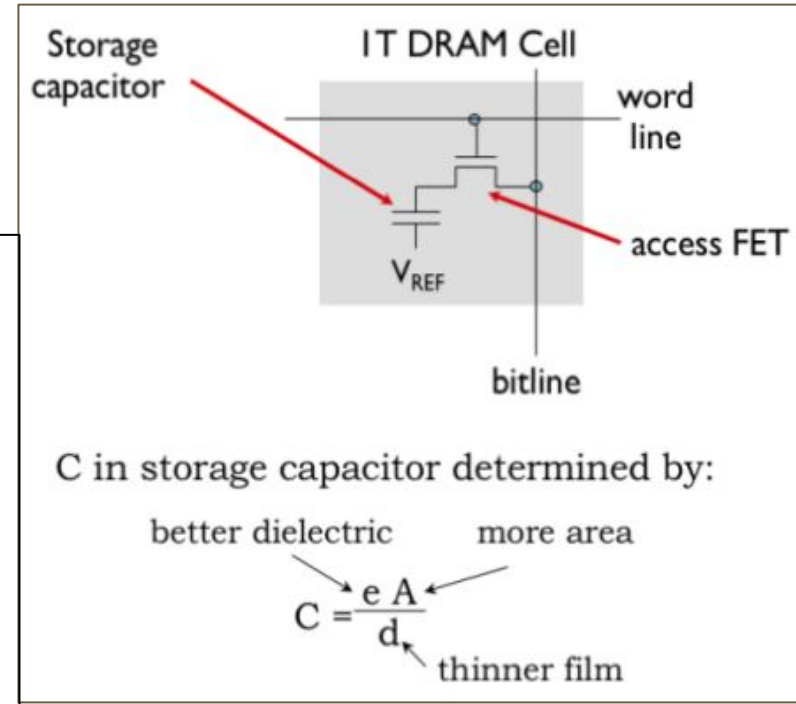
DRAM

- **DRAM** (Dynamic Random-Access Memory) is a volatile semiconductor memory used for temporary data storage in most digital systems.
- Dynamic refers to this need for periodic refresh to keep the data intact.
- Random access means that each bit of the data could be read/written independently.
- 1T1C design :
 - It stores each bit of data using a single capacitor and one access transistor.
- DRAM is denser, compact, and low-cost, ideal for large memories.
- DRAM is 20-times less area/bit than an SRAM bit cell.
- It uses charge-based storage, capacitor leaks charge, must be refreshed periodically (every 64 ms).
 - When there is Charge in capacitor = 1 (High)
 - When there is NO Charge in capacitor = 0 (LOW)

DRAM

- There are two kinds of power: Static and Dynamic Power. In DRAM:
 - Static power consumption = Low.
 - Dynamic power consumption = High (Due to refreshing).
- Commonly used in:
 - Main RAM in laptops, desktops, servers
 - Works in modules: DDR4, DDR5
 - Graphics DRAM (GDDR)
 - Mobile DRAM (LPDDR)

- The basic DRAM memory cell consists of:
 - 1 Transistor (Access switch) :
 - To select which bits will be affected by read and write operations.
 - 1 Capacitor (Storage element) :
 - If the capacitor $>$ threshold, we're storing a 1 bit, otherwise 0.



DRAM Write Operation

- DRAM write operations are straightforward: simply turn on the access FET with the wordline and charge or discharge the storage capacitor through the bitline.
- Let say we want to write '1'
- Set Write Data on Bit Line: Drive Bitline(BL) to VDD or GND, for writing '1', Bit Line is driven to VDD (logic high).
- Activate Word Line: Row decoder selects the required row WL goes high. Access transistor (T) turns ON. Bit Line and Capacitor are now connected.
- Data Stored in Capacitor: If $BL = VDD$ capacitor charges up stores logic '1' If $BL = 0$ capacitor discharges stores logic '0' This state remains as long as leakage doesn't drain the capacitor (hence refresh needed).
- Deactivate Word Line: WL is pulled low Access transistor turns OFF Cell is isolated with data stored as charge or no charge

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Thank You !

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